EXPERIMENT 12: IMPLEMENTING SECURITY FEATURES AND GOVERNANCE / ADMINISTRATION PRACTICES IN ZOHO CREATOR

Aim

To configure, enforce, and test enterprise security controls, administrative governance policies, audit trails, IP whitelisting, PII data encryption, session limits, and Zia AI-assisted threat detection in Zoho Creator for the Automated Corporate Talent Acquisition and Interview Tracking System.

Theory

Enterprise cloud platform governance requires robust security mechanisms to protect sensitive candidate data, ensure regulatory compliance (such as GDPR, SOC 2, and HIPAA), and safeguard system infrastructure against unauthorized access or data exfiltration. Zoho Creator combines platform-level administrative controls, network restrictions, encryption standards, and AI security monitoring.

Key security, governance, and administration concepts implemented in this module include:

- **IP Whitelisting & Network Restrictions:** Restricting workspace access to authorized corporate IP addresses and CIDR blocks prevents unauthorized logins from external, untrusted locations.
- **PII Field Encryption & Masking at Rest:** Sensitive candidate Personally Identifiable Information (PII)—such as Government National ID, SSN, Passport Number, and Salary details—is encrypted using AES-256 at rest, preventing exposure across unauthorized database exports or API dumps.
- **Audit Logging & Compliance Trails:** Comprehensive Audit Logs capture every system event (login attempts, record reads, modifications, exports, schema updates, and deletion requests) with timestamp, user ID, IP address, and old vs. new field value differentials for forensic compliance auditing.
- **Session Management & Authentication Policies:** Multi-Factor Authentication (MFA), SAML-based Single Sign-On (SSO), forced session timeouts, and concurrent login limits eliminate credential sharing and session hijacking risks.
- **Zia AI Governance & Anomaly Detection:** Zia AI continuously scans platform usage patterns to detect anomalous activities—such as mass data export attempts outside business hours, suspicious bulk record deletions, or unusual login locations—automatically triggering security alerts and session locks.

> **□ ZIA AI GOVERNANCE & THREAT DETECTION CONSOLE**
>
> - **Zia Anomalous Data Export Guard:** Monitors report export volumes; if a recruiter attempts to export >100 candidate records at once, Zia halts the operation and triggers an admin alert.
> - **Zia PII Masking & Data Leakage Scanner:** Scans custom Deluge scripts and email body templates to prevent accidental logging or emailing of unencrypted PII data.
> - **Zia Mass Deletion Interceptor:** Intercepts bulk record deletion commands in candidate databases, requiring multi-factor administrative re-authentication.

- **Zia Compliance Audit Summarizer:** Generates monthly regulatory compliance summaries (GDPR / ISO 27001 audit reports) detailing data access patterns and administrative privilege changes.

Procedure

1. Log in to Zoho Creator Admin Console as the Enterprise System Administrator.

2. Navigate to Setup -> Security & Governance -> IP Restrictions and add authorized corporate IP ranges to the IP Whitelist.

3. Configure Authentication Policies under Zoho Directory, enabling mandatory Multi-Factor Authentication (MFA) and setting session inactivity timeouts to 15 minutes.

4. Open the Candidate Application form builder, select sensitive fields (Government ID, Expected Salary, Passport Number), and enable Encrypt Field at Rest (AES-256).

5. Configure Audit Trail Settings across candidate, evaluation, and user forms to log all Create, Read, Update, and Delete (CRUD) events with field-level delta tracking.

6. Set up Export Restrictions under Application Governance, disabling raw CSV/XLSX export permissions for operational recruiter profiles while maintaining view access.

7. Navigate to Zia AI Security Settings and enable Zia Anomaly Detection and Threat Radar for data export and login activities.

8. Write a Deluge Security Script (`On User Input` / `On Validate`) to enforce custom IP restriction rules and log unauthorized form access attempts into a custom Security_Audit_Log form.

9. Implement a Deluge Rate-Limiting workflow that monitors candidate record view frequencies per user, locking sessions if suspicious activity is detected.

10. Perform Security Testing by simulating logins from unapproved IP addresses, attempting unauthorized PII field views, and executing bulk exports to verify blocking and alert generation.

11. Inspect the Audit Log Console and Zia Security Incident Report to confirm that all security events were accurately logged.

ZIA AI PRIVACY & ACCESS CONTROL ADVISORY

- **Zia Dynamic Privilege Reclamation:** Detects inactive user accounts (no login > 30 days) and automatically downgrades permissions to read-only until admin review.
- **Zia IP Anomaly Detection:** Correlates login locations and device fingerprints, flagging impossible travel speed logins (e.g., login from Chennai followed by login from London within 10 minutes).
- **Zia GDPR Right-to-be-Forgotten Automation:** Automates compliance erasure workflows, ensuring candidate deletion requests purge associated subform records, logs, and CRM links securely.

Sample Deluge Security & Administrative Governance Scripts

Below are reference Deluge security code implementations configured and tested during the experiment:

1. Custom IP Address Verification & Access Enforcement Script

```
// On Load Event - Candidate Application Form
// Verifies client IP address against authorized corporate network list
clientIP = zoho.ipaddress;
allowedIPList = {"192.168.1.100", "10.0.4.15", "203.0.113.50"};

if (!allowedIPList.contains(clientIP) && zoho.loginuser != zoho.adminuserid)
{
    // Log security violation
    insert into Security_Audit_Log
    [
        User_Email = zoho.loginuser,
        Attempted_IP = clientIP,
        Event_Type = "UNAUTHORIZED_IP_ACCESS",
        Timestamp = zoho.currenttime
    ];
    alert "Access Restricted: Your current IP address (" + clientIP + ") is not
authorized to access Candidate PII data.";
    cancel submit;
}
```

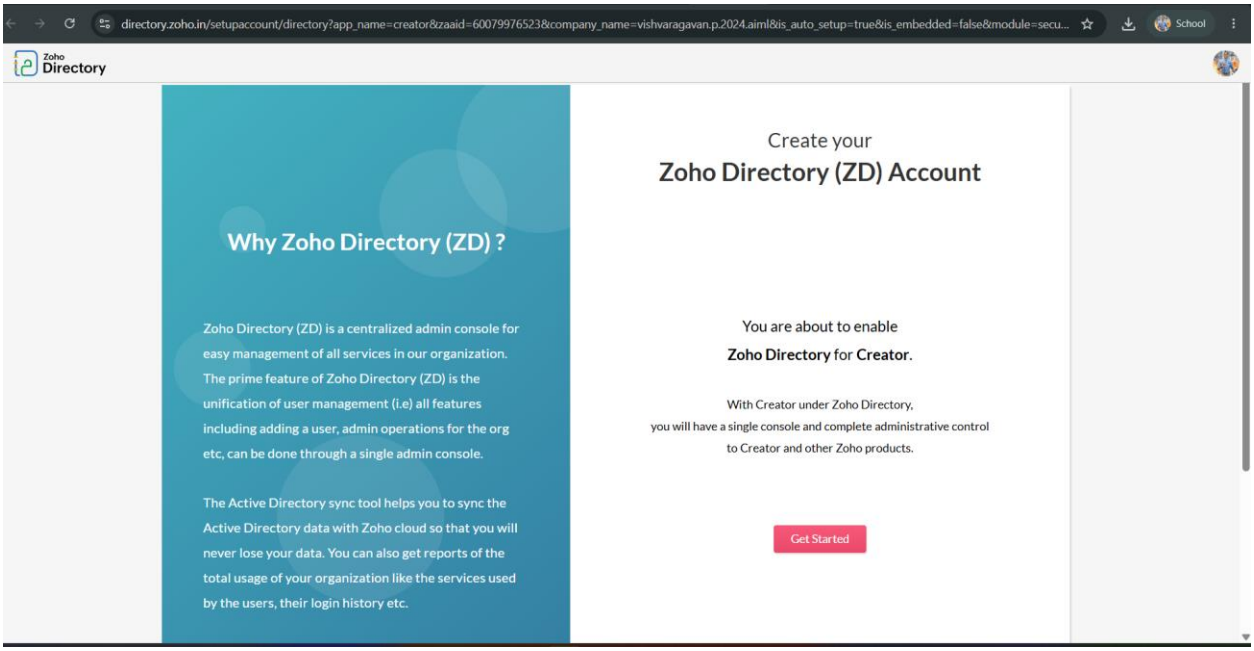
2. Automated Audit Trail Logging Trigger for Candidate Record Edits

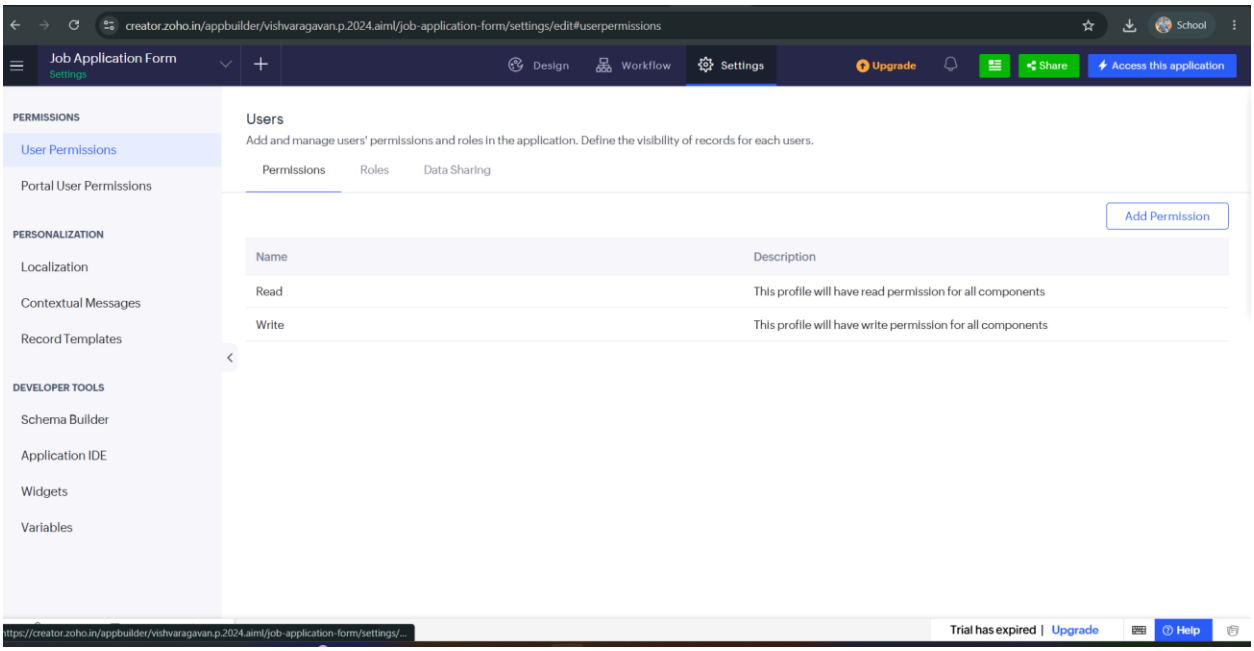
```
// On Success Event (Update) - Candidate Application Form
// Creates detailed audit trail entry with before/after field values
if (oldvalue.Application_Status != input.Application_Status)
{
    insert into System_Audit_Trail
    [
        Record_ID = input.ID,
        Module_Name = "Candidate_Application",
        Modified_By = zoho.loginuser,
        Field_Name = "Application_Status",
        Old_Value = oldvalue.Application_Status,
        New_Value = input.Application_Status,
        Change_Timestamp = zoho.currenttime,
        User_IP = zoho.ipaddress
    ];
    info "Audit log generated for status update on record: " + input.ID;
}
```

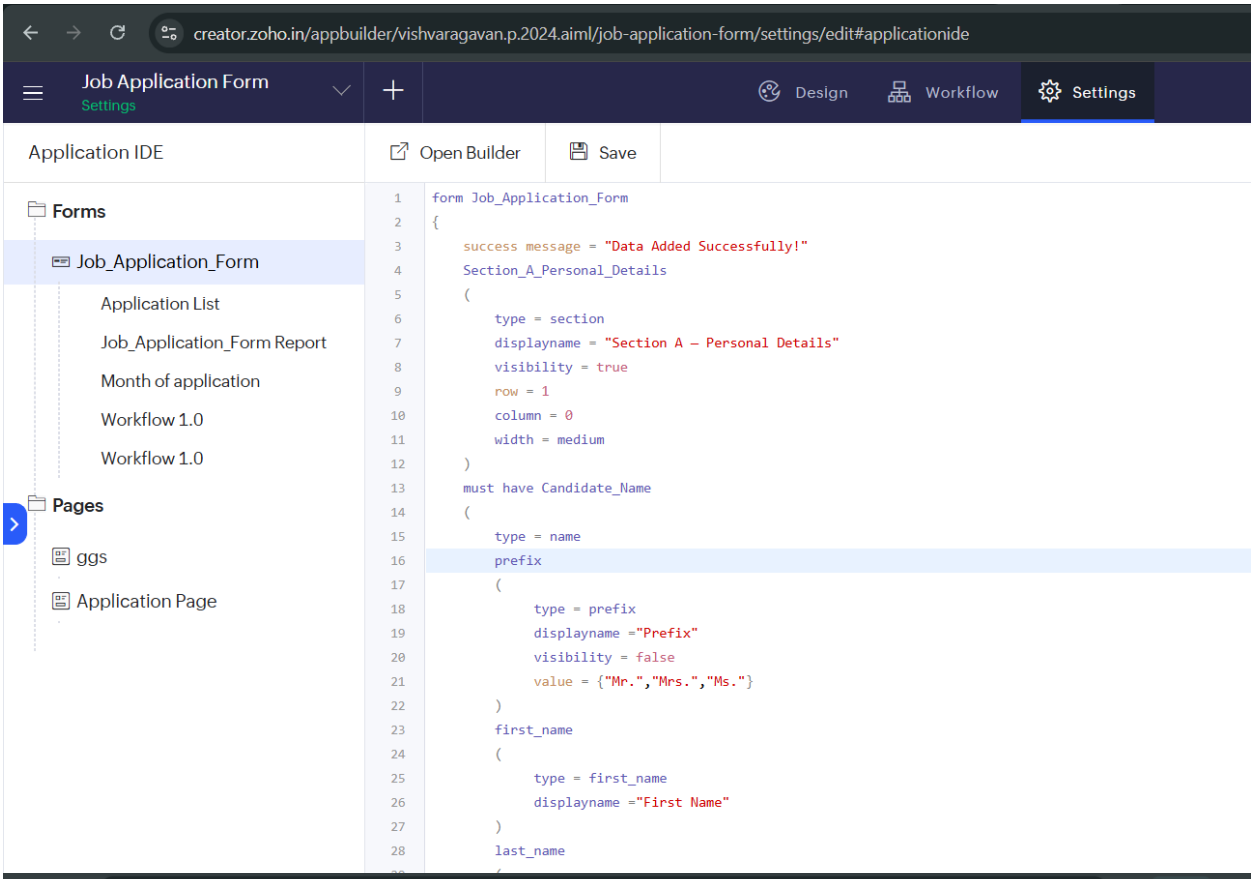
3. Mass Export Prevention & Rate Limiting Script (Zia AI Integration)

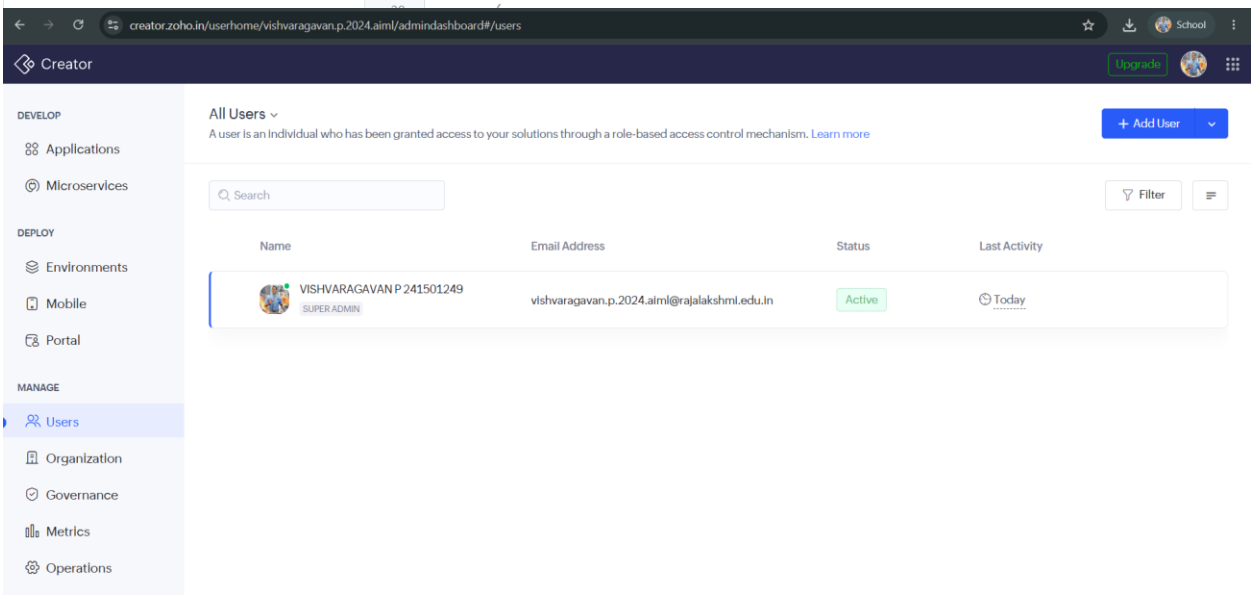
```
// On Custom Action / Report Query - Bulk Export Trigger
```

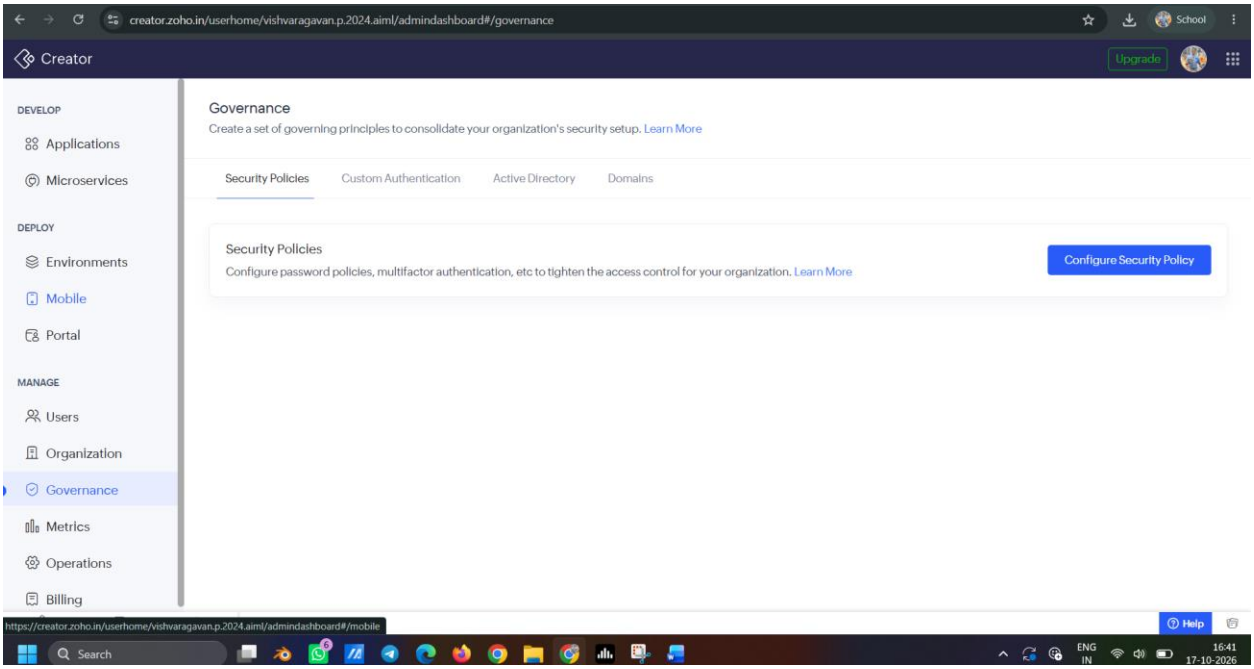
```
// Rate limits mass data retrieval to prevent data theft
requestedCount = input.Record_Count;
if (requestedCount > 50 && zoho.loginuser != zoho.adminuserid)
{
    // Trigger Zia Security Alert
    sendmail
    [
        From    : zoho.adminuserid
        To      : "security-admin@company.com"
        Subject : "SECURITY ALERT: High-Volume Candidate Export Attempted"
        Message : "User " + zoho.loginuser + " attempted to export " +
requestedCount + " candidate records from IP: " + zoho.ipaddress
    ];
    alert "Export Blocked: Bulk exports exceeding 50 records require HR Director
approval.";
    cancel submit;
}
```

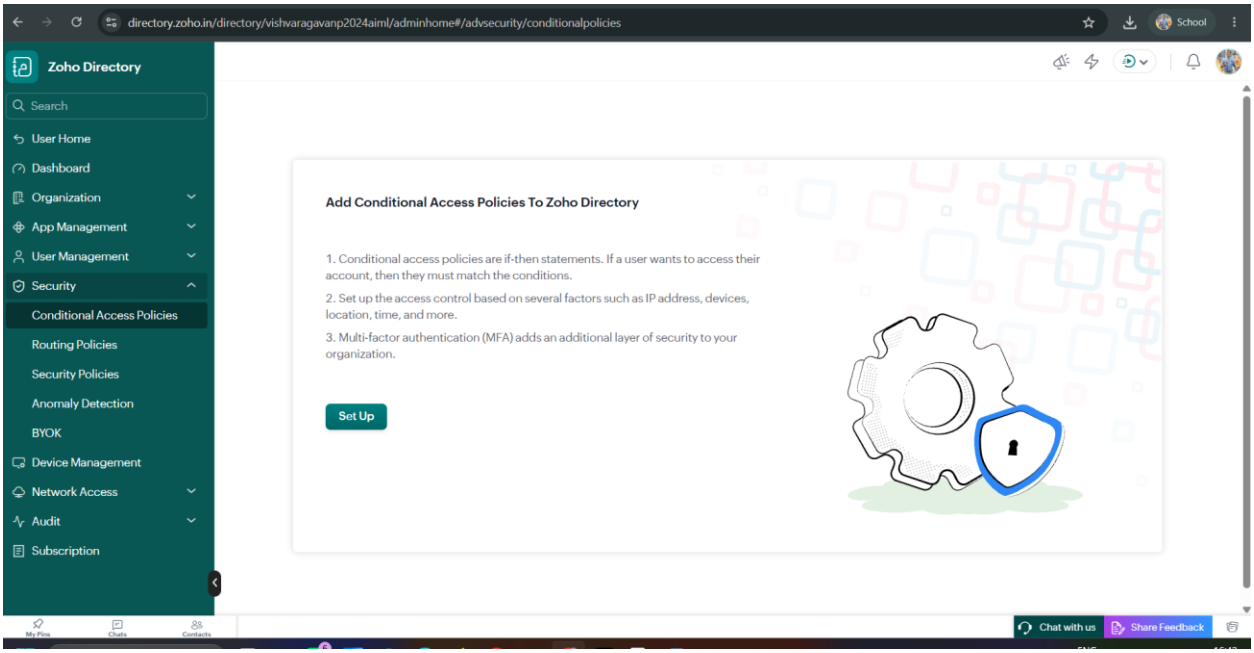



creator.zoho.in/appbuilder/vishvaragavan.p.2024.aiml/job-application-form/settings/edit#userpermissions
School
Job Application Form
Settings
Design
Workflow
Settings
Upgrade
Share
Access this application
PERMISSIONS
User Permissions
Portal User Permissions
PERSONALIZATION
Localization
Contextual Messages
Record Templates
DEVELOPER TOOLS
Schema Builder
Application IDE
Widgets
Variables
Users
Add and manage users' permissions and roles in the application. Define the visibility of records for each users.
Permissions
Roles
Data Sharing
Add Permission
Name
Description
Read
This profile will have read permission for all components
Write
This profile will have write permission for all components
https://creator.zoho.in/appbuilder/vishvaragavan.p.2024.aiml/job-application-form/settings/...
Trial has expired | Upgrade
Help


creator.zoho.in/appbuilder/vishvaragavan.p.2024.aiml/job-application-form/settings/edit#applicationide
Job Application Form
Settings
Design
Workflow
Settings
Application IDE
Open Builder
Save
Forms
Job_Application_Form
Application List
Job_Application_Form Report
Month of application
Workflow 1.0
Workflow 1.0
Pages
ggs
Application Page
form Job_Application_Form
{
success message = "Data Added Successfully!"
Section_A_Personal_Details
(
type = section
displayname = "Section A – Personal Details"
visibility = true
row = 1
column = 0
width = medium
)
must have Candidate_Name
(
type = name
prefix
(
type = prefix
displayname ="Prefix"
visibility = false
value = {"Mr.","Mrs.","Ms."}
)
first_name
(
type = first_name
displayname ="First Name"
)
last_name


creator.zoho.in/userhome/vishvaragavan.p.2024.aiml/admindashboard#/users
School
Creator
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Result

Thus, enterprise security features and administrative governance practices—including IP whitelisting, PII data encryption at rest, comprehensive audit logging, export rate-limiting, and Zia AI threat detection—were successfully implemented, configured, and verified in Zoho Creator for the Automated Corporate Talent Acquisition and Interview Tracking System.